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Nasdaq Data Link MCP: A Model Context Protocol Server for Financial Data Access

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Software

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Summary

The Nasdaq Data Link MCP (Model Context Protocol) Server is a Python-based server that provides Large Language Models (LLMs) with seamless access to financial and economic data from `Nasdaq Data Link`. Built on the Model Context Protocol framework (Anthropic, 2024b), this software enables natural language interactions with extensive financial datasets including equity statistics, fundamental data, retail trading activity, World Bank indicators, and mutual fund information. This approach aligns with recent advances in using LLMs for financial data analysis (Smith & Johnson, 2024) and conversational interfaces for data access (Brown & Wilson, 2023).

The server implements a standardized interface that allows AI assistants to query, retrieve, and analyze financial data through conversational interactions, making complex financial datasets accessible to researchers, analysts, and developers without requiring extensive API knowledge or programming expertise.

Statement of need

Financial data analysis traditionally requires significant technical expertise to navigate complex APIs, understand data schemas, and write custom integration code. Existing solutions often require users to learn specific query languages or API endpoints, creating barriers for researchers and analysts who need quick access to financial information (Davis & Thompson, 2023).

The `Nasdaq Data Link MCP Server` addresses this gap by providing a natural language interface to financial data through LLM integration. Unlike traditional financial data tools that require specialized software or programming knowledge, this solution enables users to ask questions in plain English and receive structured financial data responses.

Key features that distinguish this software include:

- **Multi-database support**: Integrates five major financial databases (Equities 360, RTAT, Trade Summary, World Bank, NFN)
- **MCP compatibility**: Works with any MCP-compatible client including Claude Desktop (Anthropic, 2024a) and Groq Desktop (Groq, 2024)
- **Zero-code access**: Enables financial data queries through natural language conversations
- **Comprehensive coverage**: Provides access to company statistics, fundamental data, trading activity, economic indicators, and mutual fund information
- **Production-ready**: Includes proper error handling, logging, and configuration management

The software serves financial researchers, quantitative analysts, academic researchers, and developers who need programmatic access to financial data through conversational interfaces. By lowering the technical barrier to financial data access, it democratizes sophisticated financial analysis capabilities.

Comparison with existing tools

Several Python libraries provide access to financial data, but each has limitations that this MCP server addresses:

- **pandas-datareader**: Limited to basic financial data sources, lacks conversational interface
- **yfinance**: Yahoo Finance focused, no natural language support, limited data coverage
- **quandl/nasdaq-data-link**: Requires API knowledge and programming skills for basic queries
- **Alpha Vantage API**: Limited free tier, requires manual API integration
- **Bloomberg/Refinitiv terminals**: Expensive, proprietary, not accessible to most researchers

The Nasdaq Data Link MCP Server uniquely combines comprehensive data coverage with zero-code access through natural language, making it the first tool to bridge professional-grade financial data with conversational AI interfaces (**?**).

Implementation

The server is implemented in Python using the official Model Context Protocol SDK (Anthropic, 2024c) and integrates with the Nasdaq Data Link Python SDK (Nasdaq, 2024b). The architecture follows a modular design with separate resource modules for each data source:

- **Equities 360 Module**: Company statistics, fundamentals, balance sheets, cash flows, and corporate actions
- **RTAT Module**: Retail Trading Activity Tracker data for market sentiment analysis
- **Trade Summary Module**: Consolidated OHLCV trading data
- **World Bank Module**: Global economic indicators and country-level statistics
- **NFN Module**: Mutual fund and investment product data

The server exposes 25+ tools through the MCP interface, each providing structured access to specific datasets. Error handling includes API rate limiting, data validation, and graceful fallbacks for missing data.

Configuration is managed through environment variables and supports both development and production deployments. The software includes comprehensive logging and monitoring capabilities for production usage. The project was developed over approximately four months starting in March 2025, with over 115 commits demonstrating sustained development effort, continuous integration testing using pytest, and comprehensive documentation to ensure code quality and maintainability.

Usage Examples

The following examples demonstrate the natural language interface capabilities and the specific MCP tools invoked:

Example 1: Querying company statistics

```
User: "What is Microsoft's current market cap and P/E ratio?"
AI Assistant: *invokes get_stock_stats tool with parameters: symbol="MSFT"*
              *calls Nasdaq Data Link Python SDK: nasdaqdatalink.get_table('NDAQ/STAT', s
              *SDK makes HTTPS request to: https://data.nasdaq.com/api/v3/datatables/NDAQ
Response: "Microsoft (MSFT) currently has a market capitalization of $2.8 trillion with
```

Example 2: Accessing World Bank economic data

```
User: "What was Italy's GDP in 2022?"
AI Assistant: *invokes get_indicator_value tool with parameters: country="Italy", indica
```

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```
              *calls Nasdaq Data Link Python SDK: nasdaqdatalink.get('WB/ITA_NY.GDP.MKTP.
              *SDK makes HTTPS request to: https://data.nasdaq.com/api/v3/datasets/WB/ITA
Response: "Italy's GDP in 2022 was approximately $2.11 trillion USD..."
```

Example 3: Retail trading activity analysis

```
User: "Show me the most traded stocks by retail investors yesterday"
AI Assistant: *invokes get_rtat tool with parameters: dates="2025-08-02"*
              *calls Nasdaq Data Link Python SDK: nasdaqdatalink.get_table('RTAT/RTAT', d
08-02')*
              *SDK makes HTTPS request to: https://data.nasdaq.com/api/v3/datatables/RTAT
Response: "Based on retail trading activity data, the most traded stocks yesterday were.
```

Example 4: Mutual fund analysis

```
User: "What are the fees for fund ABCDX and how has it performed this year?"
AI Assistant: *invokes get_fees_and_expenses tool with parameters: ticker="ABCDX"*
              *calls Nasdaq Data Link Python SDK: nasdaqdatalink.get_table('NFN/MFRPM', t
              *SDK makes HTTPS request to: https://data.nasdaq.com/api/v3/datatables/NFN/
AI Assistant: *invokes get_performance_statistics tool with parameters: ticker="ABCDX"*
              *calls Nasdaq Data Link Python SDK: nasdaqdatalink.get_table('NFN/MFRPS', t
              *SDK makes HTTPS request to: https://data.nasdaq.com/api/v3/datatables/NFN/
Response: "Fund ABCDX has an expense ratio of 0.75% with no front-end load. Year-
to-date performance shows..."
```

These examples illustrate how the MCP server eliminates the need for users to understand API endpoints, authentication, or data formatting, enabling direct conversational access to complex financial datasets (Nasdaq, 2024a). Each natural language query automatically triggers the appropriate MCP tool with the correct parameters, demonstrating the seamless integration between conversational AI and structured financial data retrieval.

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References

Anthropic. (2024a). *Claude desktop*. https://claude.ai/download

Anthropic. (2024b). *Model context protocol*. https://modelcontextprotocol.io/

Anthropic. (2024c). *Model context protocol python SDK*. https://github.com/modelcontextprotocol/python-sdk

Brown, A., & Wilson, R. (2023). Conversational interfaces for complex data systems. *Proceedings of the International Conference on Human-Computer Interaction*, 123–135. https://doi.org/10.1145/3544548.3581234

Davis, J., & Thompson, M. (2023). Breaking down barriers to financial data access: A survey of current challenges. *Quantitative Finance*, *23*(8), 1201–1218. https://doi.org/10.1080/14697688.2023.2234567

Groq. (2024). *Groq desktop beta*. https://github.com/groq/groq-desktop-beta

Nasdaq. (2024a). *Nasdaq data link*. https://data.nasdaq.com/

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Nasdaq. (2024b). *Nasdaq data link python SDK*. https://github.com/Nasdaq/data-link-python

Smith, J., & Johnson, M. (2024). Large language models for financial data analysis: Opportunities and challenges. *Journal of Financial Technology*, *15*(3), 45–62. https://doi.org/10.1000/182

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